

UNIREN 1% w/w Gel**TOPICAL ANTI-INFLAMMATORY/ ANTI-RHEUMATIC**

Each 100 g contains:
Diclofenac diethylammonium eq. to
Diclofenac sodium 1 g (1% w/w)

❖ PRODUCT DESCRIPTION:

Colorless, transparent, smooth, homogeneous gel with characteristic odor

❖ MECHANISM OF ACTION:**Pharmacology**

The mechanism by which Diclofenac affects actinic keratoses is unknown. Like other Non-Steroidal Anti-Inflammatory Drugs (NSAIDs), Diclofenac inhibits the actions of prostaglandins.

Pharmacokinetics

Absorption: Approximately 10% of the Diclofenac dose was absorbed systemically following topical application four times daily for seven days.

Distribution: Volume of distribution (Vol_d) is 550 mL per kg following oral administration of Diclofenac.

Protein binding: Very high (99%)

Biotransformation: The metabolism of topically administered Diclofenac, thought to be similar to that following oral administration, likely results in several phenolic metabolites, two of which are biologically active. Following topical administration, however, Diclofenac and its metabolites are present in the plasma in amounts too small to allow quantification of specific metabolites.

Time to peak concentration: 4.5 ± 8 hours following topical application.

Peak plasma concentration: 4 ± 5 nanograms per mL following topical application.

Elimination: Diclofenac and its metabolites are excreted mainly in the urine after oral dosing.

❖ INDICATION:

Treatment of:

- Localized forms of soft-tissue rheumatism, e.g. tendovaginitis, shoulder-hand syndrome, bursitis, and peri-arthritis.
- Localized rheumatic disease, e.g. osteoarthritis of the spine and peripheral joints;
- Post-traumatic inflammation of the tendons, ligaments, muscles, and joints, e.g. due to sprains, strains, or bruises.

❖ DOSAGE AND ADMINISTRATION:

Topical

Depending on the size of the painful area to be treated, 2-4 g of UNIREN Gel (cherry to walnut sized amount, which is enough for an area of about 400-800 cm²) should be applied 3-4 times daily to the affected sites and gently rubbed in. The duration of treatment depends on the indication and the patient's response. It is advisable to review the treatment after two weeks.

After assessing the risk/ benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

❖ WARNING AND PRECAUTION:

Concomitant use of oral NSAIDs should be cautioned as the incidence of untoward effects, particularly systemic side effects, may increase.

UNIREN Gel should not be co-administered with other products containing Diclofenac.

UNIREN Gel should be applied only to intact, non-diseased skin and not to skin wounds or open injuries.

It should not be used with occlusion.

It should not be allowed to come into contact with the eyes or mucous membranes, and should never be taken by mouth. Some possibility of gastrointestinal bleeding in those with a significant history of this condition has been reported in isolated cases.

Risk of Gastrointestinal Ulceration, Bleeding and Perforation with NSAIDs

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Cardiovascular Thrombotic Events

Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAIDs, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration. There is no consistent evidence that the concurrent use of Aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAIDs use.

Hypertension

NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAIDs treatment and at regular intervals thereafter.

Product	Code No.	Dimension	Packaging Type	Thickness
UNIREN GEL	IUMY 0102	W 15 x L 25 cm.	Wood Free Paper (กระดาษปลอดกาว)	60 g (0.08 mm.)
Designed by: (PDM, ASPD, PDCM, PDCA, PDCM, PDCA) Checked by: (PDM/ASPD) Approved by: (MD - Only Pattern and Coding)				
PLC: (Date) LRA: (Date) IRA: (Date) QCC-PM: (Date) CUSTOMER/SALES DEPT: (Date)				

PM Specification

Heart Failure

Fluid retention and edema have been observed in some patients taking NSAIDs, therefore caution is advised in patients with fluid retention or heart failure.

Gastrointestinal Events

All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patient about signs and symptoms of serious gastrointestinal toxicity.

The concurrent use of Aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Severe Skin Reactions

NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, Toxic Epidermal Necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

❖ **PREGNANCY AND LACTATION:**

Since no experience has been acquired with UNIREN Gel in pregnancy or lactation, it is not recommended for use in these circumstances. During the last trimester of pregnancy the use of prostaglandin synthetase inhibitors may result in premature closure of ductus arteriosus, or in uterine inertia.

❖ **SIDE EFFECT:**Local reactions:

UNIREN Gel is usually well tolerated.

Occasional: Allergic or non-allergic contact dermatitis (with symptoms and signs such as itching, reddening, edema, papules, vesicles, bullae or scalling of skin).

Systemic reactions:

Isolated cases: Generalized skin rash, hypersensitivity reactions (e.g. asthmatic attacks, angioedema), photosensitivity reactions. Patients should be warned against excessive exposure to sunlight in order to reduce the incidence of photosensitivity.

General:

Systemic absorption of UNIREN Gel is low compared with plasma levels obtained following administration of oral forms of Diclofenac and the likelihood of systemic effects occurring with topical Diclofenac is small compared with the frequency of side effects associated with oral Diclofenac.

However, where UNIREN Gel is applied to a relatively large area of skin and over a prolonged period, the possibility of systemic side effects cannot be completely excluded. Asthma has been rarely reported in patients using topical NSAIDs preparations.

❖ **CONTRAINDICATION:**

UNIREN Gel is contraindicated in patients with a known hypersensitivity to Diclofenac, methylparaben, menthol, peppermint oil, butylated hydroxytoluene, isopropyl alcohol, propylene glycol, carbomer 940 and trolamine.

Because of cross-reactions, the gel should not be used by patients who have experienced hypersensitivity reactions such as symptoms of asthma, allergic rhinitis or urticaria, to acetylsalicylic acid or other non-steroidal anti-inflammatory agents. The use of UNIREN Gel is contraindicated during the last trimester of pregnancy.

❖ **DRUG INTERACTION:**

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs), other (potential for increased adverse effects).

❖ **OVERDOSE AND TREATMENT:**

An overdose of topical Diclofenac Gel is extremely unlikely due to the low systemic absorption of this dosage form. Oral ingestion of topical Diclofenac sodium gel resulting in significant systemic side effects, although not reported to have occurred to date, may be treated as follows:

Treatment of overdose

To decrease absorption: Induction of emesis and/ or gastric lavage should be initiated.

Oral administration of activated charcoal may help reduce absorption.

To enhance elimination: Forced diuresis theoretically may increase urinary excretion.

Note: The use of dialysis or hemoperfusion to enhance elimination of highly protein-bound Diclofenac is unproven.

Specific treatment: Symptomatic treatment for potential complications (such as renal failure, seizures, gastrointestinal irritation, and respiratory depression) may be given.

Supportive care: Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

❖ **STORAGE:**

Store at temperature of not more than 30°C.

❖ **DOSAGE FORM AND PACKAGING AVAILABLE:**

Gel, Tube (Aluminum) 25 g

❖ **DATE OF REVISION:**

June 28, 2023

Manufactured by:

UNISON LABORATORIES CO., LTD.

39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

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